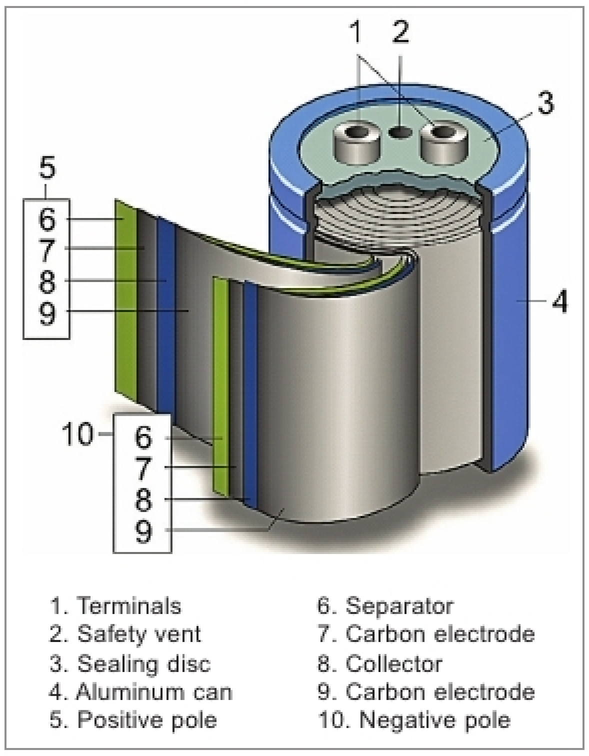

Fig. 3: Exploded view of a wound supercapacitor

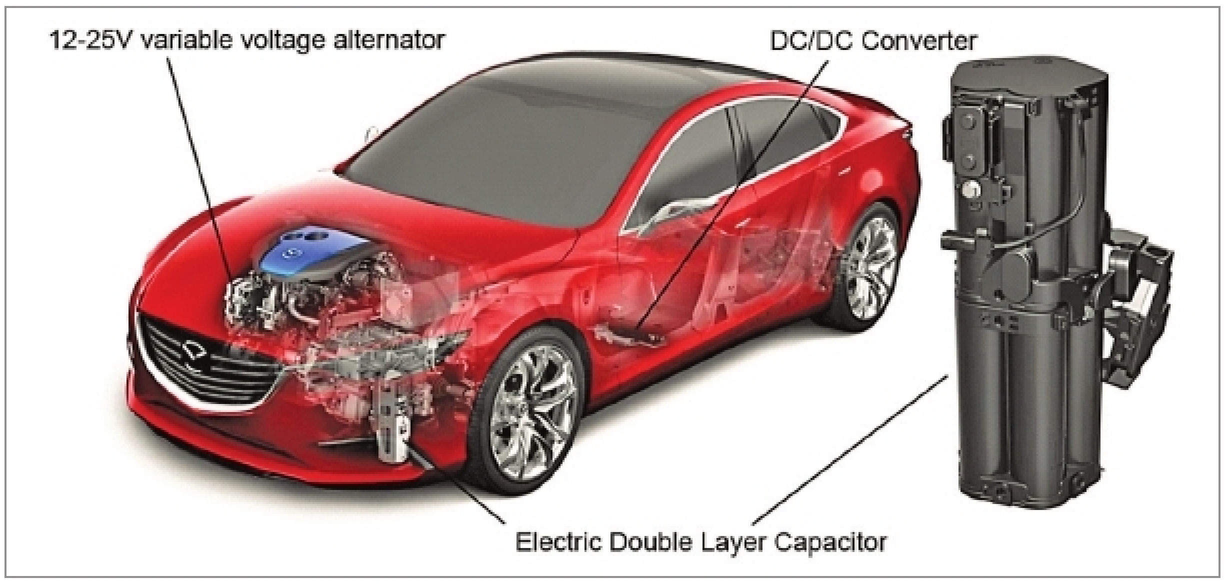

Fig. 4: A hybrid motor using supercapacitor

Fig. 5: Capa bus

Virtually all supercapacitor manufacturers use activated carbon made from coconut shells as the active component of the electrode. During processing, the ash is removed and kept below one per cent, iron is kept below 100ppm and halogens are removed in order to have an extended cycle of operation. High porosity of this material can provide an electrode area up to 3000 sq. metre/gm. It can yield capacitance of 145 F/gm. Active material represents half the total cost of a supercapacitor. As the cost of activated carbon is cheaper, about ₹ 1050/kg, it is being widely used.

A carbon nanotube (carbon molecules with a cylindrical nanostructure) is another promising active material, but is quite costly to manufacture. Carbon nanotubes cost about ₹ 3500/kg, but have double the energy density, and therefore are used for special-purpose supercapacitors. The nanotube surface provides greater wettability and has high conductivity. It can yield capacitance up to 180 F/gm.

Graphene, a two-dimensional carbon monolayer, is another promising candidate whose cost varies from ₹ 350 to ₹ 2800/kg. But its fluffiness limits the energy density of a supercapacitor and can't be competitive, unless a way is found to tightly pack it.

For pseudocapacitors based on Redox, Ruthenium dioxide yielding capacitance of 750 F/gm, Manganese dioxide, which gives capacitance of 350 F/gm, IrO_2, etc are typically used as electrodes.

Electrolytes. Electrolytes consist of solvent and dissolved chemicals that dissociate into positive cations and negative anions. Aqueous acidic or alkaline electrolytes are used in supercapacitors with low specific energy and high specific power. Electrolytes with organic solvents are more expensive but can give a higher capacitor voltage of 2.7V.

Uses of supercapacitors

Vehicles. Supercapacitors are used in vehicles for regenerative braking. They have much higher power densities than batteries and can charge and discharge more rapidly. They also store much more energy as compared to conventional electrolytic capacitors. This makes them more suitable for capturing kinetic energy as a vehicle slows and release it for bursts of acceleration.

They don't have the ability to store large amounts of energy, but they are high powered devices, primarily because the electrons can move back and forth very quickly. Supercapacitors can provide extra power of up to 34hp and can be used to smooth out acceleration, bridging the gaps in power delivery that occurs when the mechanical transmission changes gear. This removes any jerkiness that would occur during a shift. As supercapacitors can charge and discharge symmetrically and in seconds, they make for an incredibly efficient means of providing a short but strong electric assist in a vehicle while driving. The supercapacitor is discharged during acceleration and recharged during braking, which means that a hybrid motor or an e-motor never lacks the power it needs during driving, ultimately increasing the efficiency of the vehicle.

As shown in Fig. 4, the system generally uses a 12-25V variable energy alternator that generates electricity during regenerative braking at up to 25V before sending it to a low-resistance electric double-layer capacitor that can be fully charged in seconds. A DC-DC converter then steps down the electricity from 25V to 12V before distributing it to the vehicle's electrical system.

Capa bus. A capacitor vehicle or capa vehicle, as shown in Fig. 5, is a traction vehicle that uses supercapacitors to store electricity. In general, supercapacitors can store about 5 per cent of the energy that lithium-ion rechargeable batteries can. This limitation restricts the range of driving to a couple of kms per charge. But they can be charged in a matter of seconds.